User Guide · Demo 8 Hull Generator

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Demo 8 makes a ship hull from six numbers, C_B, LCB, L, B, T and D, by transforming the KCS parent hull in two steps (Lackenby, then a stretch). It returns the hull's offsets, its hydrostatics, its resistance, a lines plan, a 3D view and, optionally, a STEP file for CAD. One hull takes less than a second, so it can sit inside an optimization loop. The method is explained in Reading SP05; this guide only explains how to use the code.

Open it. Course site › Groups › Specialization resources › SP05 › *Run*: *Demo 8 · Hull Form Variation from a Parent (notebook)*. It opens in Google Colab. First *File › Save a copy in Drive*, so that your changes are kept, then run the cells from the top with Shift+Enter (or *Runtime › Run all*). Nothing needs installing, except for the STEP file (below). The notebook loads the KCS offsets from the course site by itself.

Quick start: one hull in five lines

Run sections 1–8 once (they define the functions), then, in a new cell:

```
hull = make_hull(CB=0.690, LCB=-1.0, L=200.0, B=30.0, T=10.5, D=16.0)   # lengths in m,
LCB in % L, + forward
summary(hull)                       # c_F, c_A and the hydrostatics: V, C_B, C_P, C_M,
C_WP, LCB, LWL, KB, BM, KM
lines_plan(hull)                    # profile, half-breadth plan and body plan
view_3d(hull)                       # interactive 3D view
export_offsets(hull, "my_hull.csv")  # the offsets, same layout as KCS_offsets.csv
```

The example values are not a project design. For the example, `summary` prints c_F +0.134, c_A +0.082, V 43,470 m³, C_B 0.6900, C_P 0.7004, C_M 0.9851, C_{WP} 0.8542, LCB −1.00 %, KB 5.67 m, BM 7.97 m and KM 13.64 m. Its resistance:

```
r = resistance(hull["h"], hull["B"], hull["T"], 20.0)   # speed in kn
print(r["RT"], r["PE"])                                 # R_T in kN, P_E in kW: 1,176
kN and 12,098 kW
```

Map of the notebook

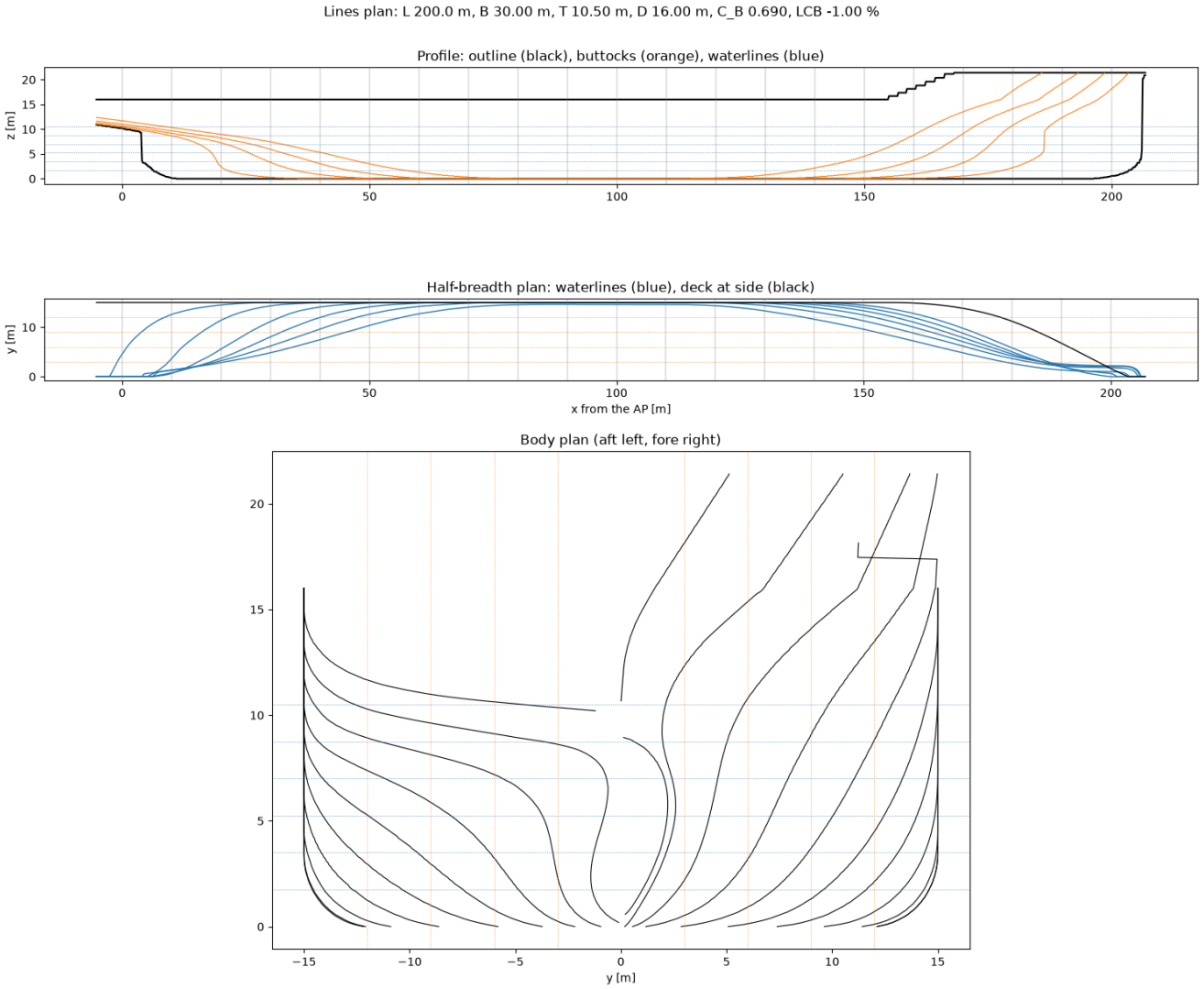
Sections 1–7 teach the method step by step; section 8 packs it into functions for your own work; section 9 is the optional STEP export. Change only the lines marked `# INPUT`.

Section	What it does	What you can change (`# INPUT`)
1 · The parent	Loads the KCS offsets from the course site	–
2 · Hydrostatics	Defines `hydrostatics`; checks the KCS against its published particulars	–
3 · Lackenby	Defines `lackenby` and `solve_lackenby`; shifts the KCS to a target C_B and LCB	`CB_target`, `LCB_target`
4 · Stretch	Defines `stretch`; scales the hull to L, B, T and D; shows that the coefficients do not change	`L`, `B`, `T`, `D`
5 · Look at the result	Sectional area curves and body plans, parent against variant	–
6 · Resistance	Holtrop–Mennen with the hull's own coefficients; the check against the KCS model test	`V_kn`
7 · A family of hulls	Several C_B at the same L, B, T: volume, R_T and P_E	the C_B range
8 · Toolkit	`make_hull`, `hull_y`, `export_offsets`, `summary`, `lines_plan`, `view_3d`	your six hull variables
9 · STEP file (optional)	Installs OpenCascade and defines `export_step`	–
10 · Limits	What to state in your report	–

Function reference

Units: metres, m², m³, knots, kN, kW. x is measured from the AP, positive forward; z from the baseline, up; y is the half-breadth, to starboard.

Function	Inputs	Returns
`make_hull(CB, LCB, L, B, T, D, warn=True)`	C_B; LCB in % L, + forward of midship; L (Lpp), B, T, D	A dictionary `hull`: offsets `x`, `y`, `z`; the inputs; `cF`, `cA`; hydrostatics `h` (`V`, `CB`, `CP`, `CM`, `CWP`, `LCB`, `LWL`, `KB`, `BM`, `KM`, `ABT`, `hB`); `warnings`
`summary(hull)`	a hull	Prints the factors and the hydrostatics
`hull_y(hull, x, z)`	x and z, numbers or arrays	The half-breadth y; 0 outside the hull
`export_offsets(hull, filename)`	a file name, e.g. `"my_hull.csv"`	A CSV with one row per station (`x_m`) and one column per waterline (`z=...`)
`resistance(h, B, T, V_kn)`	the hydrostatics `hull["h"]`, B, T, speed	Holtrop–Mennen: `Fn`, `RT`, `RFk`, `RW`, `RB`, `RA`, `PE`, and the propulsion factors `w`, `t`, `etaR`, `etaH`
`lines_plan(hull, n_wl=6, n_bt=4, n_st=21)`	numbers of waterlines, buttocks and stations	Profile, half-breadth plan and body plan
`view_3d(hull)`	a hull	Interactive 3D view: drag to rotate, scroll to zoom
`export_step(hull, filename, n_st=161, n_g=100)`	a file name, e.g. `"my_hull.step"`; numbers of stations and of points around each section	A STEP file of the hull surface, both sides (section 9)



Using it in your design model

A loop over candidates. `make_hull` takes about 0.1 s, so it can be called for every candidate of a sweep or an optimizer. Collect what you need from each hull:

```
results = []
for CB in [0.66, 0.68, 0.70]:
    for L in [140.0, 150.0, 160.0]:
        hull = make_hull(CB, -1.0, L, B=25.0, T=9.0, D=14.0, warn=False)
        r = resistance(hull["h"], hull["B"], hull["T"], 18.5)
        results.append((CB, L, hull["h"]["V"], hull["h"]["KM"], r["PE"],
hull["warnings"]))
```

The half-breadth for the container count. The bay-by-bay count needs the hull's half-breadth at the ends of each bay and at the bottom and top of each tier. `hull_y` gives it at any point. A sketch of the idea, for one tier of one bay; the clearances and the cell-guide allowance are yours to set and justify:

```
w_box = 2.438                       # container width [m], ISO 668
def rows_in(hull, x_aft, x_fwd, z_low, z_high, side_clear, pitch):
    """Containers side by side between x_aft and x_fwd, z_low and z_high: the narrowest
corner governs."""
    y_min = min(hull_y(hull, x, z) for x in (x_aft, x_fwd) for z in (z_low, z_high))
    width = 2 * (y_min - side_clear)
    return max(int((width + pitch - w_box) // pitch), 0)
```

The function above checks only the hull: the double hull, the hatch openings and the engine room limit the holds too. Write your own count with them, and check it against Lecture 3.2.

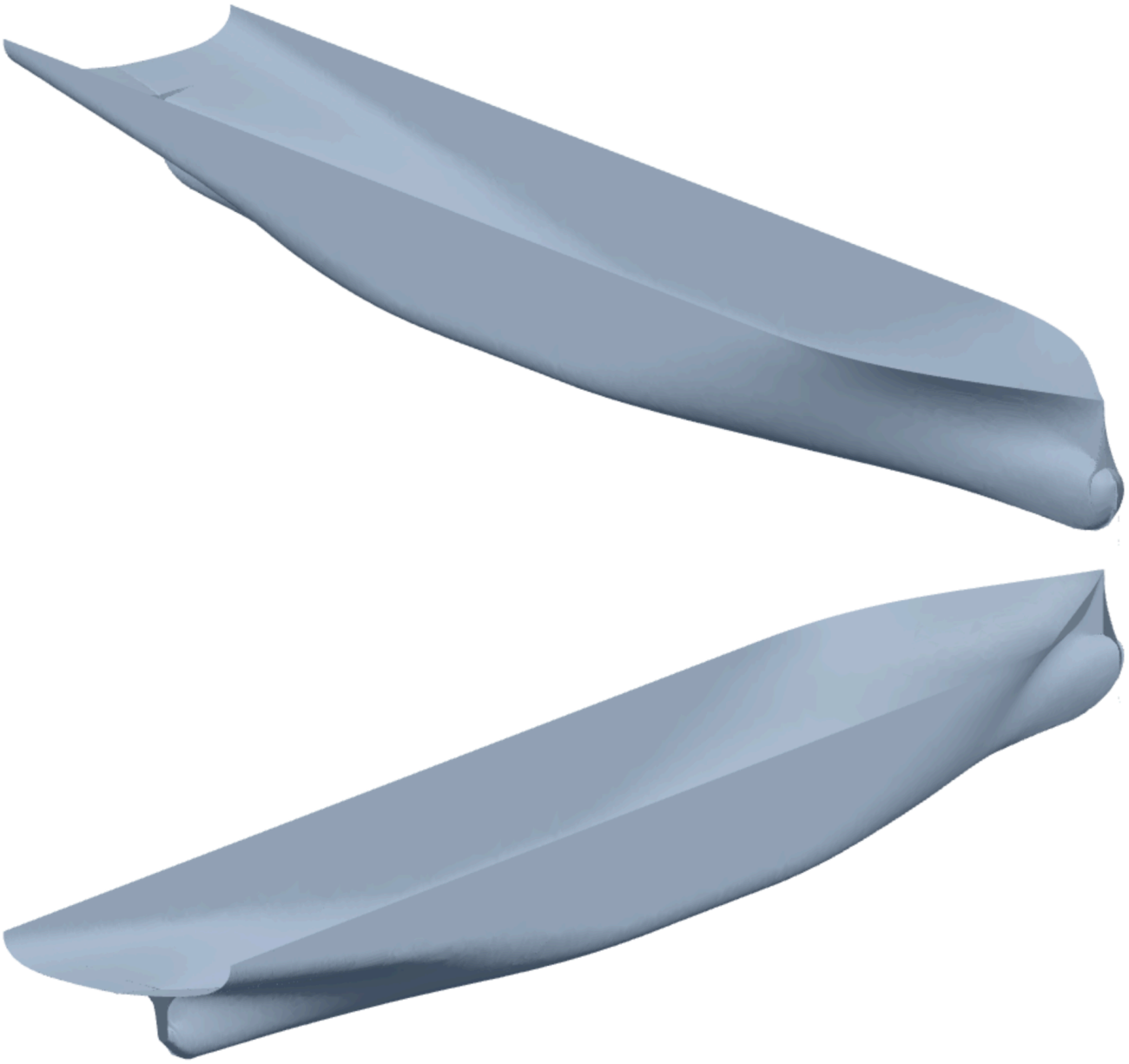
STEP file for CAD

`export_step` writes the hull surface to a STEP file that CAD programs open (FreeCAD, Rhino, SolidWorks, Siemens NX and others), in metres.

1. Run the install cell of section 9 once per session: `!pip -q install cadquery-ocp==7.8.1.1.post1` (about a minute).
2. Run the cell that defines `export_step`, then call it on your hull:

```
export_step(hull, "my_hull.step")                  # n_st=161 stations, n_g=100 points
around each section
from google.colab import files; files.download("my_hull.step")
```

What you get. Two faces, starboard and port, from the keel up to the deck at side, D. The surface passes exactly through the hull at the stations and points, and is straight between them; more points give a closer surface and a larger file (about 3 MB for the defaults). Not included: the forecastle above D, the deck and the transom, which are left open, and the rudder and propeller. The figure below is the STEP file of the quick-start example, read back with OpenCascade and rendered.



A ten-minute tour for class

1. **Open and copy** the notebook in Colab; *Runtime › Run all*. Point at the parent check of section 2: the offsets give the published volume, C_B and LCB.
2. **Lackenby** (section 3): set `CB_target = 0.70`, run; show c_F and c_A, the check $c + (1 - c)\, C_P$ for each half, and the new parallel body.
3. **Stretch** (section 4): change `L`, `B`, `T` and `D`; show that C_B, C_M, C_P, C_{WP} and LCB do not change, and that the volume ratio equals $s_x s_y s_z$.
4. **Look** (section 5): sectional area curves and body plans, parent against variant.
5. **Power** (sections 6–7): P_E of the hull, the +5.7 % check against the KCS model test, and the family of C_B with its rising power.
6. **Toolkit** (section 8): `make_hull` with six numbers, `lines_plan`, and `view_3d`, rotated live.
7. **Out of range:** `make_hull(0.61, -1.0, 150, 25, 9, 14)` prints a warning: C_B is below the valid range.
8. **CAD** (section 9, optional): install, export the STEP file and open it in a CAD program.

Warnings, limits and troubleshooting

Warnings printed by `make_hull`. Outside C_B 0.64–0.71, L/B 5.5–7.5 or B/T 2.5–3.5 the hull is still made, but it is an extrapolation of the KCS: say so in your report. A "target not reached" warning means Lackenby could not reach the C_B and LCB asked for: move them towards the parent's values (C_B 0.653, LCB −1.50 %).

Limits of the hull (Reading SP05): the bulb and the stern are stretched, not redesigned; C_M stays at the parent's 0.985; the resistance is Holtrop–Mennen, about 6 % above the model test on the parent.

Problem	Cause and fix
`NameError: name 'make_hull' is not defined`	The cells above were not run: *Runtime › Run all*, or run from the top
The offsets do not load	No internet in the session: download `KCS_offsets.csv` from the course site and upload it to the Colab files panel; the notebook reads the local copy first
`ModuleNotFoundError: No module named 'OCP'`	Run the install cell of section 9; if it still fails, *Runtime › Restart session* and run again
The 3D view is blank	Run the cell again; plotly needs the browser to allow scripts in the output
A strange hull	Check the units (m, % L) and the sign of LCB (+ forward); check the warnings